TRIBHUVAN UNIVERSITY

CENTRAL DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Object Oriented Software Engineering

Assignment No-2

Submitted By:

Name: Yujan Basnet

Roll No: 18

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Submitted to:

Prof. Dr. Subarna Shakya

Difference between Hierarchical Object-Oriented Design (HOOD), Object Modeling Technique (OMT) and Responsibility Driven Design (RDD)

Features	HOOD	OMT	RDD
Main Focus	Top-down hierarchical decomposition.	Static structure and relationships.	Roles and behavior of objects.
Key Concepts	Hierarchy of objects	Three models (object, dynamic, functional).	Responsibilities, contracts, client-server model.
Key Developers	European Space Agency	James Rumbaugh	Rebecca Wirfs-Brock
Design Approach	Break complex systems into simpler components with parent-child relationships.	Use three interconnected models: object, dynamic, and functional.	Define objects by their responsibilities and collaborations.
Diagrams Used	Hierarchical diagrams showing containment.	Class diagrams, state diagrams, data flow diagrams.	CRC (Class-Responsibility-Collaboration) cards.
Strengths	Clear organization, manageable complexity through top-down modular decomposition.	Comprehensive modeling framework that integrates structure, behavior, and data flow.	Creates flexible and maintainable systems by focusing on well-defined object responsibilities and collaborations.
Abstraction Level	System-level abstraction	Mixed abstraction (class to system)	Object-level abstraction
Use Case	Real-time, embedded systems	Influenced UML development and General OO modeling	Agile and domain-driven design